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1.1 Derivative in Computer Vision

Image derivatives are the mathematical foundation of both edge and corner detection, as they capture local intensity variations that correspond to meaningful structural features. A 2D image can be treated as 2D discrete function $I(x, y)$. The derivatives are approximated via finite differences or convolution with derivative kernels.

The first order derivatives explains the rate of change of intensity in an image, representing the edges as local maximum $\nabla I(\frac{\partial I}{\partial x}), \frac{\partial I}{\partial y}$. The second order derivative describes the rate of change of the gradient, and the edges are represented as zero crossing point (as in LoG) $\nabla^2 I(\frac{\partial^2 I}{\partial x^2}), \frac{\partial^2 I}{\partial y^2}$. However the LoG function is high computational complex, and it is approximated by DoG for computational efficiency.

1.2 Gradient magnitude and direction

From the first-order derivatives I_x and I_y , — typically computed via the Sobel operator, a separable filter combining a derivative kernel with a smoothing kernel — two quantities are derived:

- Magnitude: $|\nabla I| = \sqrt{I_x^2 + I_y^2}$, which encodes the edge strength (high values represents a sharp transition)
- Direction: $\theta = \arctan(\frac{I_y}{I_x})$, which encodes the edge orientation (perpendicular to edge contours)

These quantities are crucial to apply NMS, since the edge direction is used to determine the orientation of the local neighborhood for non-maximum suppression, ensuring that only the most significant edge pixels are retained and edge magnitude is used to threshold the edges, discarding weak edges that are likely to be noise.

1.3 Second order derivative for Intensity transitions

The laplacian (LoG) responds to intensity transitions with a zero crossing at the edge location, remarking edge boundary with high spatial accuracy. However, its main challenge is noise since it is amplified by the function, solved by a pre-smoothing by a Gaussian filter. Two main strategies are used to overcome the noise sensitivity:

- Gaussian smoothing, before differentiation
- Medial filtering, non linear strategy for salt and pepper noise

1.4 Corner detection

Corner detection relies on image derivatives and a second-moment matrix approach (structure tensor). The second moment matrix H over a window W is defined as:

$$H = \sum_{(x,y) \in W} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

To analyze the local intensity variation, we study the eigenvalues λ_1 and λ_2 of H , which correspond to the magnitudes of maximum and minimum intensity change:

- $\lambda_1 \approx \lambda_2 \approx 0$ (Flat region): The error is low in all directions; intensity is nearly constant.
- $\lambda_1 \gg \lambda_2 \approx 0$ (Edge): The error is high in one direction but low along the edge.
- $\lambda_1 \approx \lambda_2 \gg 0$ (Corner): The error is high in all directions, representing a point of distinct intersection.

Since calculating eigenvalues for every pixel is computationally expensive, the Harris detector introduces a response function R that approximates their behavior without explicit eigenvalue decomposition:

$$R = \det(H) - k \cdot (\text{trace}(H))^2 = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

where k is an empirical constant (typically 0.04 – 0.06). The value of R indicates the "cornerness":

- R large and positive $\implies$ Corner.

- R large and negative $\implies$ Edge.
- $|R|$ small $\implies$ Flat region.

Properties of the Harris detector:

- **Equivariant** to translation and rotation (eigenvalues and shape are preserved under rotation).
- **Invariant** to additive brightness shifts ($I' = I + b$) because derivatives eliminate constant offsets.
- **Not scale-invariant**, as a physical corner may appear as an edge or a flat region when zoomed in.



2.1 Detecting keyframes in Video

The all approach can be divided in 3 main features, to detect all the different situation in a video sequence.

- Macro (global changes)
- Micro changes
- Optical flow magnitude for movements

The all algorithm is completed by using a minimum gap constraint to avoid considering a number k of successive frames, to not face the redundancy problem.

2.2 Macro

To have a general and comprehensive understanding of the scene, the color histogram in HSV can be analyzed. This approach is not complex and allows to detect scene interrupts and macro changes in the video sequence. The two histograms are compared with Chi square distance.

2.3 Micro

It may happen that the color histogram in two consequent scenes is similar, but the objects displacement is completely different. This issue may be overcome by the use of an edge ratio system: a canny edge detector algorithm is used to find all edges in a scene and the amount of edges in two scenes is compared finding a change ratio.

2.4 Optical Flow

Finally, to detect the beginning of actions, the optical flow magnitude can be used $M = \frac{1}{N} \sum \sqrt{u^2 + v^2}$. LK comparing algorithm is used to compare consecutive frames and it can detect in two scenes some movements (mouth moving while talking), where the scenes appear similar to the Macro analyzer.



3.1 Padding Strategies

Padding is used to allow the filter to shift on all pixels of the image, including corners and edges. Many different strategies of padding exist:

- Zero and Constant padding: The borders are filled with 0s or constants, creating artificial edges on the borders
- Mirror: Reflects the inner pixel values excluding the border pixels
- Symmetric: Reflects the inner pixel values including border pixels
- Clamp: Replicates the most external pixel constantly

3.2 Effect on Output Dimensions (Valid vs. Same vs. Full)

The size of the output image depends on the input size N , kernel size K , padding size P , and stride S :

$$\text{dim}_{\text{out}} = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

- **Valid** ($P = 0$): No padding is applied. The output is smaller than the input ($\text{dim}_{\text{out}} = N - K + 1$ for $S = 1$).

- **Same** ($P = \lfloor K/2 \rfloor$): Padding is chosen such that the output has the exact same spatial dimensions as the input (for $S = 1$).
- **Full** ($P = K - 1$): Padding is added such that every element of the kernel visits every input pixel. The output is larger than the input ($\dim_{\text{out}} = N + K - 1$).

3.3 Impact on Edge Detection Near Boundaries

Edge detection relies on computing spatial derivatives (gradients):

- **Zero and Constant Padding** ($c \neq 0$): Introduce a sharp, artificial step-discontinuity between real pixel intensities and the constant padding value. Derivative filters (like Sobel) mistake this jump for a strong edge, generating **false/spurious edges** along the image borders.
- **Clamp / Replicate Padding**: Forces the gradient at the boundary to zero by replicating pixel values. While it avoids artificial step-edges, it incorrectly ****suppresses legitimate real edges**** crossing near the boundary.
- **Mirror and Symmetric Padding**: Preserve smooth gradient continuity without sharp steps. They are the optimal strategies for edge detection near image boundaries as they ****correctly preserve edge responses**** without creating false positives or suppressing true borders.



4.1 Estimating optical flow

To estimate optical flow, the Lk algorithm is employed. It is based on three main assumption:

- Brightness constancy: $I(x, y, t) = I(x + u\delta t, y + v\delta t, t + \delta t)$
- Small motion
- Spatial coherence

Starting from the brightness constancy, by applying Taylor approximation, we can derive the optical flow constraint equation $I_x u + I_y v + I_t = 0$. Since this single-pixel equation is underconstrained (1 equation for 2 unknowns), Lucas-Kanade introduces the spatial coherence assumption, yielding N^2 equations for

an $N \times N$ patch. $Ax = b$ is so formed as an overdetermined system solvable by Least Squares. The Solution is the eigenvector corresponding to the smallest eigenvalue. This is possible only if $A^T A$ is invertible and well conditioned. Indeed, by the $A^T A$ eigenvalues we can understand the optical flow, as for the harris corner detector:

- $\lambda_1 \approx \lambda_2 \approx 0$ (Flat region): No reliable optical flow can be estimated.
- $\lambda_1 \lambda_2 \approx 0$ (Edge): Aperture problem; only the normal flow (perpendicular to the edge) can be estimated.
- $\lambda_1 \approx \lambda_2 \gg 0$ (Corner / Texture): Optical flow can be estimated reliably in all directions.

The **aperture problem** occurs in the edge case, stating that observing a moving edge through a small local aperture only allows estimating the motion component perpendicular to the edge, while the parallel (tangent) component remains completely ambiguous. Lucas-Kanade mitigates this by pooling information from a spatial neighborhood (patches) rather than relying on isolated pixels.



5.1 Role of DFT in Image Filtering

The Discrete Fourier Transform (DFT) converts an image from the spatial domain into the frequency domain, decomposing it into a weighted sum of 2D sinusoids. Its primary role in image filtering relies on the **Convolution Theorem**, which states that a spatial convolution $g * h$ is equivalent to an element-wise multiplication in the frequency domain:

$$\mathcal{F}\{g * h\} = \mathcal{F}\{g\} \cdot \mathcal{F}\{h\}$$

Instead of performing costly sliding-window spatial convolutions—whose computational complexity grows quadratically with the kernel size as $O(L^2 N^2)$ —filtering can be efficiently executed in $O(N^2 \log N)$ using the Fast Fourier Transform (FFT). By the Convolution Theorem, this global approach completely eliminates the need for local window movement: we transform the entire image via FFT, transform the kernel—**zero-padded to the image dimensions to avoid circular**

convolution artifacts—via FFT, perform a single global element-wise multiplication of their spectra, and apply the **Inverse FFT (IFFT)** to return to the spatial domain. This method is especially advantageous when applying very large filter kernels.

5.2 Information Provided by the DFT

The DFT transforms an image into a complex-valued 2D spectrum $F(u, v) = R(u, v) + iI(u, v)$, providing two fundamental types of information:

- **Amplitude Spectrum (Magnitude):** $|F(u, v)| = \sqrt{R^2 + I^2}$, representing *how much energy* of a specific spatial frequency is present in the image.
 - Low frequencies (near the center, DC component) correspond to smooth, large-scale structures and background.
 - High frequencies (far from the center) correspond to fine details, sharp edges, and rapid intensity variations (textures).
- **Phase Spectrum:** $\phi(u, v) = \text{atan2}(I, R)$, which encodes the *spatial alignment* of those frequency components, defining the exact geometric location of edges, lines, and structures in the image space. **Phase is perceptually more important than amplitude:** in phase-swapping experiments, reconstructing an image using the amplitude of Image A and the phase of Image B yields an image whose structural layout and recognizable shapes belong strictly to Image B.

5.3 Filtering Examples: Low-Pass and High-Pass

- **Low-Pass Filtering:** Used for smoothing or blurring. We multiply the frequency spectrum by a mask (such as an ideal cutoff mask beyond a radius r or a smooth 2D Gaussian function) that preserves low frequencies near the center and attenuates high frequencies. *Example:* Removing high-frequency Gaussian sensor noise or smoothing background details.
- **High-Pass Filtering:** Used for sharpening or edge detection. We multiply the frequency spectrum by a mask that attenuates low frequencies at the center and preserves high frequencies in the outer peripheral regions. *Example:* Isolating abrupt intensity transitions to highlight edges and fine textures.



6.1 SIFT Detection

The Scale-Invariant Feature Transform (SIFT) detects stable keypoints in scale-space by constructing a **Difference of Gaussians (DoG)** pyramid across octaves:

$$\text{DoG}(x, y, \sigma) = G(x, y, k\sigma) - G(x, y, \sigma)$$

A pixel is selected as a keypoint if it is a local extremum compared to its **26 neighbors** in the 3D scale-space (8 spatial, 9 in the scale above, 9 below). Unstable points—specifically low-contrast points and edge responses (filtered via a 2D Hessian ratio test)—are discarded. Finally, a **canonical orientation** is assigned based on the peak of a local gradient orientation histogram to ensure rotation invariance.

6.2 SIFT Descriptor

To represent the local appearance around a keypoint:

- A 16×16 pixel window is extracted at the keypoint's characteristic scale.
- The coordinate system and gradients are rotated according to the keypoint's canonical orientation (achieving rotation invariance).
- The window is divided into a 4×4 grid of cells (16 cells total).
- An **8-bin orientation histogram** is computed for each cell.
- Concatenating all 16 histograms yields a single **128-dimensional vector** ($16 \times 8 = 128$).
- The vector is normalized to unit length (L_2 norm) for illumination invariance.

6.3 SIFT Properties

The SIFT algorithm is robust and invariant to:

- **Scale:** Via multi-scale DoG extrema detection.
- **Rotation:** Via canonical orientation alignment.
- **Illuminazione:** Via L_2 vector normalization.



7.1 SURF Detection

SURF (Speeded-Up Robust Features) is a faster alternative to SIFT, designed to address high computational complexity while maintaining comparable matching quality.

Instead of using Difference of Gaussians (DoG), SURF is a **Hessian-based** detector that identifies blobs at locations where the determinant of the Hessian matrix H is maximal:

$$H(x, \sigma) = \begin{bmatrix} L_{xx}(x, \sigma) & L_{xy}(x, \sigma) \\ L_{xy}(x, \sigma) & L_{yy}(x, \sigma) \end{bmatrix}$$

where L_{xx} represents the convolution of the image with the second partial derivative of a Gaussian.

To achieve high computation speed, SURF approximates these Gaussian second-order derivatives using **box filters**, which can be evaluated extremely fast in constant time using **integral images** (where each location stores the sum of all pixel intensities in the rectangle from the origin). The approximate determinant is computed as:

$$\det(H_{\text{approx}}) = D_{xx}D_{yy} - (wD_{xy})^2 \quad (w = 0.9)$$

7.2 SURF Descriptor

The SURF descriptor is designed to be compact and fast to compute:

- **Orientation assignment:** Computes Haar wavelet responses in the x and y directions within a circular neighborhood around the interest point, weighted by a Gaussian, to find a dominant orientation (ensuring rotation invariance).
- **Feature vector:** A square window of size $20s \times 20s$ (where s is the scale) is aligned with the dominant orientation and divided into a 4×4 grid of cells (16 cells total).
- For each cell, the sum of Haar wavelet responses and their absolute values are computed: $\sum dx, \sum dy, \sum |dx|, \sum |dy|$.
- Concatenating these 4 values for all 16 cells creates a **64-dimensional descriptor vector** ($16 \times 4 = 64$), making it computationally lighter and more compact than SIFT's 128-D vector.



8.1 Camera Model and Projection Matrix

Recovering 3D structure from images requires modeling how 3D world points are mapped to 2D pixel coordinates. This is expressed by the 3×4 projection matrix $P = K[R \mid t]$, which unifies the intrinsic and extrinsic parameters.

8.2 Intrinsic and Extrinsic Parameters

- **Intrinsic Matrix (K):** Encodes internal optical properties, mapping 3D camera coordinates to 2D pixels using focal lengths (f_x, f_y) and the principal point (o_x, o_y) :

$$K = \begin{bmatrix} f_x & 0 & o_x \\ 0 & f_y & o_y \\ 0 & 0 & 1 \end{bmatrix}$$

- **Extrinsic Matrix ($[R \mid t]$):** Describes the camera's pose relative to the world frame, consisting of a 3×3 rotation matrix R and a 3×1 translation vector t .

8.3 Camera Calibration

Camera calibration consists of computing the transformation parameters by estimating the 3×4 matrix P . Using at least **6 point correspondences** between known 3D world points and 2D image points (since P has 11 degrees of freedom), we obtain 2 linear equations per point. This forms a homogeneous system $A\mathbf{p} = 0$, solved via SVD by finding the eigenvector corresponding to the smallest eigenvalue of $A^T A$.

Once $P = [M \mid p_4]$ is estimated, intrinsic and extrinsic parameters are decoupled using **RQ factorization**:

$$M = KR \implies R = K^{-1}M, \quad p_4 = Kt \implies t = K^{-1}p_4$$

8.4 Triangulation

Once the camera matrices (P, P') and corresponding 2D points (x, x') are known from two calibrated views, the 3D position of a scene point X is recovered via **triangulation**. By using the cross product to eliminate the unknown scale factor:

$$x \times (PX) = 0 \quad \text{and} \quad x' \times (P'X) = 0$$

This generates a homogeneous linear system $AX = 0$ (4×3 system), which is solved using SVD (minimizing the algebraic error) to estimate the 3D coordinates of the point. The solution considers also the noise challenge: it is possible that the two rays are skew, thus the solution involves the minimization of the consequent error.



9.1 Introduction to Epipolar Geometry and Epipoles

Epipolar geometry describes the geometric relationship between two camera views of the same 3D scene, providing constraints that simplify feature matching.

- **Baseline:** The line segment connecting the two camera optical centers (O and O').
- **Epipolar Plane:** The plane defined by a 3D point X and the two camera centers.
- **Epipoles (e, e'):** The intersection of the baseline with the image planes. e is the projection of O' onto the first image, and e' is the projection of O onto the second image. All epipolar lines converge at the epipole.
- **Epipolar Constraint:** Given a point x in the first view, its correspondence x' in the second view must lie on the corresponding epipolar line l' , reducing the search space from 2D to 1D.

9.2 Essential Matrix E and Fundamental Matrix F

- **Essential Matrix (E):** Encodes the epipolar constraint for **calibrated cameras** (K known), operating in normalized camera coordinates ($\mathbf{x}'^T E \mathbf{x} = 0$). It is defined as $E = R[t]_{\times}$, has 5 degrees of freedom (DoFs), and rank 2.
- **Fundamental Matrix (F):** The generalization of E for **uncalibrated cameras** (K unknown), operating directly in pixel coordinates ($\mathbf{p}'^T F \mathbf{p} = 0$). It maps pixels to epipolar lines ($l' = Fp$) and has 7 DoFs.
- **Algebraic Relationship:** F and E are linked through the intrinsic camera calibration matrices K and K' :

$$F = K'^{-T} E K^{-1} = K'^{-T} R[t]_{\times} K^{-1}$$

9.3 Estimation of the Fundamental Matrix

F is estimated from corresponding point matches using the **Normalized 8-Point Algorithm**:

1. **Normalization**: Translate and scale pixel coordinates (centroid at origin, average distance $\sqrt{2}$) to ensure numerical stability.
2. **Linear System**: Set up $Ah = 0$ using ≥ 8 point matches and solve via SVD (smallest singular value of $A^T A$).
3. **Rank-2 Constraint**: Enforce $\det(F) = 0$ by computing the SVD of the estimated matrix and zeroing out its smallest singular value ($\Sigma' = \text{diag}(\sigma_1, \sigma_2, 0)$).
4. **Denormalization**: Transform F back to the original coordinate system, typically combined with **RANSAC** to reject outliers.

The normalized algorithm is preferred since the normal algorithm is sensitive to noise and to avoid the poor numerical conditioning of the A matrix, which can result in high variance in the columns' values.

9.4 Relation to Calibration Parameters and Relative Poses

- **Calibration Parameters**: Intrinsic parameters (K, K') bridge pixel coordinates and normalized coordinates. Extrinsic parameters (R, t) define the relative rotation and translation between the two camera frames.
- **Recovering Relative Pose (R, t)**: Once the Essential matrix E is recovered (either directly or via $E = K'^T F K$), it can be decomposed using SVD to yield four possible solutions for the relative camera pose (R, t). The correct physical solution is uniquely determined by checking that reconstructed 3D points lie in front of both cameras (cheirality constraint).

9.5 3D Triangulation

Given known camera projection matrices (P, P') and 2D correspondences (x, x'), the 3D position of a scene point X is recovered via **triangulation**. By eliminating the unknown scale factors via cross product ($x \times PX = 0$ and $x' \times P'X = 0$), we form a homogeneous system $AX = 0$ (4×3 system). Solving this via SVD minimizes the algebraic error to estimate the 3D spatial coordinates.



10.1 Definition of Disparity

Disparity d is the horizontal displacement (difference in pixel coordinates) of the same 3D point between a left stereo image and a right stereo image:

$$d = x_L - x_R$$

For objects in front of the cameras, $x_L > x_R$, yielding a positive disparity ($d \geq 0$). It is the core geometric parameter used to infer 3D depth.

10.2 Epipolar Rectification

Epipolar rectification is a preprocessing geometric transformation that warps both stereo images using homographies so that their epipolar lines become horizontal and coincide with the image scanlines ($v_L = v_R$). This simplifies the correspondence problem: instead of searching for matching points across a 2D image plane, the search is reduced to a 1D horizontal scan along the exact same row.

10.3 Block Matching and Similarity Metrics

To compute the disparity map, local window-based matching is commonly employed:

1. For each pixel in the left image I_L , a small patch (window) W is extracted.
2. The algorithm searches for the best matching patch along the same horizontal scanline in the right image I_R .
3. The optimal displacement d is selected using a **Winner-Takes-All (WTA)** strategy—choosing the disparity that minimizes the matching cost. Common dissimilarity metrics include:

- **SAD** (Sum of Absolute Differences):

$$\text{SAD} = \sum_{(u,v) \in W} |I_L(u, v) - I_R(u + d, v)|$$

- **SSD** (Sum of Squared Differences):

$$\text{SSD} = \sum_{(u,v) \in W} (I_L(u, v) - I_R(u + d, v))^2$$

- **NCC** (Normalized Cross-Correlation).

Window Size Trade-off: A small window captures fine details but results in a noisy disparity map; a larger window is robust to noise but blurs depth discontinuities and fails at localizing thin objects.

10.4 Relationship Between Disparity and Depth

Assuming a rectified stereo rig with two parallel cameras separated by a baseline b and sharing an effective focal length f , depth Z is inversely proportional to disparity d :

$$Z = \frac{b \cdot f}{d}$$

A high disparity indicates a close object (large horizontal shift), while a low disparity indicates a distant object (small shift).

10.5 Limitations and Failure Cases

Local block matching often fails in the following scenarios:

- **Textureless regions:** Homogeneous areas (e.g., a white wall) where all patches look identical, leading to severe matching ambiguity.
- **Repeated patterns:** Repetitive textures (e.g., bricks or fences) causing multiple identical minimum-cost matches.
- **Specular/reflective surfaces:** Mirrors or shiny metal violate the brightness constancy assumption.
- **Depth discontinuities:** Pixels near object boundaries where a local window covers parts of both the foreground and background.

❖ Domande miste

11.1 Zero-Padding vs. Mirror Padding in Sobel

Why does zero-padding generate a false high-gradient boundary along the entire image perimeter when followed by the Sobel operator?

Zero-padding adds a uniform layer of zeros around the image. When a derivative-based filter like Sobel is applied, the sharp intensity transition between the actual image content and the artificial border of zeros creates a strong,

false high-gradient response along the perimeter. Differently, mirror (or reflect) padding mirrors the boundary pixel values symmetrically, smoothing the transition and preventing artificial gradient spikes.

11.2 Limits and Validity of Homography

Under what two specific geometric conditions does a Homography matrix H (3×3) accurately map points between two images? What happens if the camera undergoes a pure translation in a non-planar 3D scene?

A homography matrix H accurately maps points between two images if and only if at least one of two conditions is met: (1) the observed 3D scene is strictly planar, or (2) the camera motion is a pure rotation around its optical center ($t = 0$), regardless of the 3D structure. If the camera undergoes a pure translation (or general motion) in a non-planar 3D scene, the **parallax effect** occurs. Points at different depths move by different amounts, making a single global homography invalid. In this scenario, one must resort to Epipolar Geometry using the Fundamental (F) or Essential (E) matrix.

11.3 Coarse-to-Fine Optical Flow and Inverse Warping

How do Gaussian pyramids, inverse warping, and residual flow solve the failure of Lucas-Kanade under large object motions?

Lucas-Kanade assumes small motion (e.g., < 1 pixel). Gaussian pyramids scale down the image resolution so that large motions (e.g., 20 pixels) at the original scale become small motions at the coarsest pyramid level. Starting from the coarsest level, the optical flow is computed, and **inverse warping** is used to align the second frame to the first without leaving gaps (avoiding forward warping holes). Then, a **residual flow** is computed for the remaining motion and propagated upward through the pyramid levels, accumulating the total optical flow.

11.4 Box Filter vs. Gaussian Filter in the Frequency Domain

Why does a Box filter produce ringing artifacts near sharp edges, while a Gaussian filter does not?

A Box filter is a rectangular function in the spatial domain, whose Fourier transform is a **Sinc function** in the frequency domain. The Sinc function exhibits prominent sidelobes that cause an abrupt and imperfect frequency cutoff,

resulting in spatial ringing artifacts. Conversely, a Gaussian filter remains a Gaussian function in both spatial and frequency domains; it has no sidelobes and provides a smooth, clean attenuation of high frequencies without ringing.

11.5 Harris Corner Detector

What is the geometric intuition behind the Harris corner detector, and how are the eigenvalues of the Second Moment Matrix interpreted?

The Harris detector looks for local intensity variations by shifting a small window in all directions. It uses the Second Moment Matrix H (or M). By analyzing its eigenvalues (λ_1, λ_2):

- Both $\lambda_1, \lambda_2 \approx 0$ (Flat region): intensity is constant in all directions.
- One large, one near zero ($\lambda_1 \gg \lambda_2$ or vice versa) (Edge): intensity changes along only one direction.
- Both $\lambda_1, \lambda_2 \gg 0$ (Corner): intensity changes significantly in all directions.

11.6 Stereo Vision Baseline Trade-off

What is the engineering trade-off in stereo vision when choosing a very small baseline versus a very large baseline?

The baseline is the distance between the optical centers of the two cameras. A **narrow baseline** makes stereo matching much easier (high overlap, fewer occlusions), but depth estimation is very sensitive to noise, as small pixel disparity errors lead to huge errors in depth calculation (Z). A **wide baseline** provides high geometric precision for depth estimation, but makes stereo matching extremely difficult due to large perspective distortions and heavy occlusions.

11.7 The Aperture Problem

What is the aperture problem in optical flow estimation, and along which direction is motion ambiguous?

The aperture problem occurs when observing a moving edge through a small local window (aperture), making it impossible to determine the true 2D motion vector. Only the motion component perpendicular (normal) to the edge can be measured. The motion component **parallel to the edge** is completely ambiguous

and cannot be reliably estimated because the intensity profile along the edge is uniform.

11.8 SIFT: Scale and Rotation Invariance

How does SIFT achieve scale invariance and rotation invariance?

Scale invariance is achieved by detecting keypoints across multiple scales using a scale-space representation based on the Difference of Gaussians (DoG) pyramids, locating stable extrema across adjacent scales. **Rotation invariance** is achieved by computing a local gradient orientation histogram in the neighborhood of each keypoint and assigning a canonical orientation corresponding to the highest peak of the histogram, so that descriptors are always aligned relative to this dominant direction.

11.9 Fundamental Matrix and its Rank

What is the Fundamental Matrix (F), and why does it have a rank of 2 ($\det(F) = 0$)?

The Fundamental Matrix F (3×3) is the algebraic representation of epipolar geometry for uncalibrated cameras, mapping a point in the first image to its corresponding epipolar line in the second image ($l' = Fx$). It has a **rank of 2** ($\det(F) = 0$) because it maps a 2D point onto a 1D line (epipolar line). Mathematically, it is defined as $F = K'^{-T}[t]_{\times}RK^{-1}$, where the essential matrix component $[t]_{\times}$ is a skew-symmetric matrix of rank 2, which forces F to also be singular (rank 2).

11.10 Epipoles and Epipolar Lines

What are epipoles and epipolar lines, and how do they simplify stereo matching?

An **epipole** (e, e') is the intersection of the baseline (the line connecting the two camera centers) with the respective image planes (equivalently, the projection of one camera center onto the other image plane). An **epipolar line** is the intersection of the epipolar plane (defined by 3D point X and the two camera centers) with the image plane. The epipolar constraint simplifies stereo matching from a 2D search problem to a 1D search problem: given a point x in the first image, its corresponding point x' in the second image must lie entirely along its corresponding epipolar line l' .

11.11 RANSAC vs. Least Squares

Why do Least Squares fail in the presence of high outlier ratios, and how does RANSAC solve this?

Standard Least Squares minimizes the sum of squared residuals, which causes even a small percentage of severe outliers (e.g., 40%). **RANSAC (Random Sample Consensus)** solves this by iteratively sampling the minimum subset of points required to estimate the model, computing the transformation, and counting how many total points agree with it (the consensus set / inliers). By keeping the model with the largest consensus set over many iterations, it completely isolates and discards the outliers.

11.12 Mean vs. Median Filter for Salt-and-Pepper Noise

Why does a Mean filter blur salt-and-pepper noise while a Median filter completely eliminates it?

A Mean filter is a linear filter that averages all pixel values in a neighborhood. Since salt-and-pepper noise consists of extreme, isolated impulse values (maximum or minimum brightness), averaging them smears the noise values across neighboring pixels, causing blurring. A Median filter is a non-linear filter that replaces the pixel value with the statistical median of its neighborhood. Extreme noise impulses fall at the extreme ends of the sorted neighborhood values and are automatically ignored, completely removing the noise without blurring edges.

11.13 Phase Swapping Experiment in Fourier Domain

In a phase-swapping experiment between Image A and Image B, which image is visually recognizable in the IFFT output and why?

Image B (the one whose **Phase** was used) will be visually recognizable. While the Amplitude spectrum contains the overall energy distribution and frequency magnitudes, the **Phase spectrum** contains the precise spatial structure, alignment, and edge locations of the image. The human visual system relies overwhelmingly on phase information to perceive shapes and objects.

11.14 Background Subtraction in Outdoor Environments

What traditional (non-Deep Learning) approach is used for background subtraction in outdoor scenes with changing illumination and moving shadows?

Simple frame differencing fails due to lighting variations and shadows. Instead, traditional computer vision uses statistical background modeling such as **Gaussian Mixture Models (GMM)**, where each pixel is modeled over time by a mixture of Gaussians to account for multi-modal backgrounds (e.g., swaying tree leaves, flickering light). Alternatively, color spaces like HSV are used, separating chromaticity from intensity (V) to make the system more robust to illumination shifts and shadows.